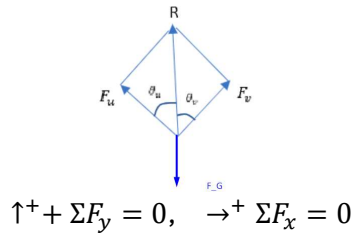


--- HW1 Content ---

Resultant Forces



$$\uparrow^+ \Sigma F_y = 0, \quad \rightarrow^+ \Sigma F_x = 0$$

Linear Solve for:

$$(1) -F_G + F_u \sin \theta + F_v \sin \theta = 0$$

$$(2) F_u \cos \theta = F_v \cos \theta$$

Robot Driving up a Slope

$$\text{Downhill Drive: } F_{net} = \frac{\tau}{R} + mg \sin(\theta) - mg \cos(\theta) C_r(v)$$

$$\text{Upward Climb: } F_{net} = \frac{\tau}{R} - mg \sin(\theta) - mg \cos(\theta) C_r(v)$$

$$F = ma, \quad t = \frac{v}{a}$$

--- HW2 Content ---

Velocity Equations in a Rotating Frame

Linear Solve for $\dot{x}$ and $\dot{y}$:

$$\dot{x}' = \dot{x}_o' - \sin(\theta) * \dot{\theta} * x + \cos(\theta) * \dot{x} - \cos \theta * \dot{\theta} * y - \sin \theta * \dot{y}$$

$$\dot{y}' = \dot{y}_o' + \cos(\theta) * \dot{\theta} * x + \sin(\theta) * \dot{x} - \sin(\theta) * \dot{\theta} * y + \cos(\theta) * \dot{y}$$

Final Position / Global-Local Displacement

$$x_A' = x_o' + x_n' + x_A * \cos(\theta) - y_A * \sin(\theta)$$

$$y_A' = y_o' + x_n' + x_A * \sin(\theta) + y_A * \cos(\theta)$$

Where $\Sigma \theta = \theta + \theta$

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

--- HW3 Content ---

Trajectory - Linear Motion

$\dot{\phi}_R$	$\dot{\phi}_L$	FWD
$\dot{\phi}_R$	$-\dot{\phi}_L$	RIGHT
$-\dot{\phi}_R$	$-\dot{\phi}_L$	REV
$-\dot{\phi}_R$	$\dot{\phi}_L$	LEFT
$\dot{\phi}_R$	$-\dot{\phi}_L$	CCW
$-\dot{\phi}_R$	$\dot{\phi}_L$	CW

Solve for $\dot{\phi}_R$, 2 cases:

$$\text{Linear Move: } \frac{\text{distance}}{\text{time}} = \text{Wheel radius } R * \dot{\phi}_R \text{ (rad)}$$

$$\text{Rotated Move: } \frac{\theta}{t} = \frac{R}{L} \theta_R \text{ (degrees, convert to rad after)}$$

Trajectory - Arc Motion

$$\dot{x} = \frac{2\pi r}{\text{time}} \text{ m/s}$$

$$\dot{\theta} = \frac{\theta_{\text{final}} - \theta_{\text{initial}}}{\text{time}} \text{ rad/s (Use Unit Circle)}$$

To Solve for Phi Dot L & R, Linear Solve the Following:

$$\dot{x} = \frac{R}{2} (\dot{\phi}_L + \dot{\phi}_R) \text{ m/s}$$

$$\dot{\theta} = \frac{R}{2L} (\dot{\phi}_R - \dot{\phi}_L) \text{ rad/s}$$

--- HW4 Content ---

Trajectory - Mathematical Function

Degrees $\rightarrow$ radians
$\times \text{ by } \frac{\pi}{180}$
Radians $\rightarrow$ degrees
$\times \text{ by } \frac{180}{\pi}$

Given $y' = \sqrt[3]{x}$ from $x'=1$ to $x'=1.5$ m:

$$\theta_{\text{final}(2)} = \arctan\left(\frac{d}{dx}(y')|_{x=1.5}\right) \text{ (degrees, convert to rad after)}$$

$$\theta_{\text{initial}(1)} = \arctan\left(\frac{d}{dx}(y')|_{x=1}\right) \text{ (degrees, convert to rad after)}$$

$$\dot{\theta} = \frac{\theta_{\text{final}} - \theta_{\text{initial}}}{\text{time}} \text{ (rad/s)}$$

$$\text{distance} = \frac{\sqrt{(y_2' - y_1')^2 + (x_2' - x_1')^2}}{2 \sin\left(\frac{|\theta_2 - \theta_1|}{2}\right)} \text{ (m)}$$

$$\dot{x} = \frac{\text{Distance} \times |\theta_2 - \theta_1|}{t} \text{ (m/s)}$$

Same steps for finding Phi Dot as in HW3 Section

